Weapon Repair and Maintenance
Ted Li

Foil Tips -- Removing and Cleaning

1. Remove as much tip tape as is necessary to gain access to the tip screws.
2. While holding the tip to prevent it flying off, remove the two tip screws using a magnetized 1.5mm or 1.8mm slotted screwdriver. Then remove the tip and the pressure spring. This step should be done over a light-colored cloth to make finding errant parts easier..
3. Using a Q-Tip with rubbing alcohol or contact cleaner, swab the inside of the barrel, being careful not to leave lint in it.
4. Reverse Step #2 to reassemble the point, making sure that the pressure spring is seated in the contact cup.
    a. On the collar-type tips (aka “French” tips), align the collar holes before inserting the tip into the barrel. Check the electrical continuity after one screw is inserted. Then insert the other screw.
    b. On the flange-type tips (aka “German” tips), after inserting and tightening the tip screws,, loosen them ¼ turn -- this allows the tip to center itself, and will ease the future removal of the tip screws.
5. Once the tip is reassembled, test the pressure spring with a 500 gram foil weight. If it fails, open the tip and replace the pressure spring, as stretching the spring is only a very temporary remedy.
6. If necessary, re-tape the foible of the blade separately from the barrel, allowing for future repairs of the tip to be done without the necessity of stripping all of the blade tape.

N.B. There should be no need to lubricate a foil point, unless it is to prevent corrosion from forming inside it. One drop of a Teflon lubricant will suffice. Oil and graphite should be avoided.
If the barrel is heavily dented or out-of-round, use either a mandrel or reamer to reform the barrel.

Epee Tips -- Removing and Adjusting

1. Follow Steps #2-#3 (listed above) in “Foil Tips -- Removing and Cleaning.” In addition to swabbing the barrel with rubbing alcohol or contact cleaner, polish the end of the contact spring (the small spring attached to the tip) with emery paper.
2. To adjust the travel on epee tips with contact springs, screw the spring clockwise to shorten it., To lengthen it, “cross-thread” it by gently pulling it while twisting it clockwise; or yank the old contact spring from the tip, and replace it with a new one.
3. With a continuity tester attached to the “A” and “B” sockets of the epee, and the pressure spring removed, place the tip in the barrel and check the *residual travel* with a 0.50mm epee gauge. With the shim inserted between the barrel and the tip there should be no continuity when the tip is depressed. When the shim is removed, and the tip depressed, there should be continuity. With the tip depressed and the blade flexed, there should be constant continuity. If there is only intermittent continuity, then the contact spring is too short.
4. To reassemble the point, insert the pressure spring and the tip, and test the pressure spring using a 750 gram epee weight. If the pressure spring will not reset the 750 gram weight, change the pressure spring. Do not merely stretch the pressure spring, as that will only temporarily fix the problem.
5. If the point passes both the weight and shim tests, reinsert the tip screws and check the continuity a final time.

N.B. There should be no need to lubricate an epee point, unless it is to prevent corrosion from forming inside it. One drop of a Teflon lubricant will suffice. Oil and graphite should be avoided.

Foil Wires - Stripping and Replacing

1. Dis-assemble the foil, removing all the tip tape. If a new blade is to be wired, skip Steps #2-#4, and thoroughly degrease the blade groove using acetone or other degreaser.
2. Strip as much of the blade wire from the blade groove as possible by pulling the wire from the tang end.
3. Remove the complete point by unscrewing it. This usually breaks the wire below the barrel. Continue stripping the wire by pulling it from the tip end.
4. Remove any remnants of wire and glue from the blade groove using a sharpened hacksaw blade, Dremel rotary tool with a thin cut-off disc, a solvent (usually acetone), or heat. Thoroughly clean the blade groove.
5. Dis-assemble the point, removing the tip screws, the tip, the pressure spring. Then, remove the insulated contact by pushing it toward the mouth of the barrel using a small screwdriver or other punch. If the contact is to be re-wired, this step must be done carefully to avoid destroying the insulating cup. Place all the removed parts on a small magnet to prevent their loss.
6. To rewire the blade, gently unwrap the new foil wire, and straighten any kinks it may have. Check to see that the pressure spring will fit into the new contact cup and that the new contact cup will fit easily into the barrel.
7. Thread about 2" of the new blade wire through the barrel, and screw the barrel onto the blade, being careful not to pinch or cut the wire in the blade groove. The use of Teflon plumbing sealant tape or medium strength Loctite on the blade threads will help prevent the barrel from loosening.
8. Tighten the barrel using a 5mm wrench or pliers, taking care not to deform the barrel or split it by over-tightening.
9. Gently pull the wire through the barrel, taking care not to strip the wire's insulation in the process.
10. Center the contact cup in the barrel. While gently pulling on the blade wire to keep it from kinking inside the barrel, use a point setter to seat the contact cup firmly at the bottom of the barrel.
11. Place the blade wire, under slight tension, into the blade groove, and wrap the excess wire around the threads of the tang. If the blade is going to be stored, a protective covering of tape can be applied to the wire wrapped around the tang.
12. Place the blade in a blade-bending jig, creating a 8"-10" arc to hold the wire at the bottom of the blade groove. Be sure the wire actually sits at the bottom of the blade groove.
13. Using the thinnest cyanoacrylate glue, begin gluing the wire by holding the blade vertically by the tip, and placing one drop of the cyanoacrylate in the blade groove, ¾" behind the point, letting it "wick" down the blade wire. Continue applying the cyanoacrylate one drop at a time, until the glue reaches the middle of the blade. Then quickly wipe the blade groove with a tissue to remove any excess glue, and to hasten the glue's setting.
14. Holding the blade by the tang, repeat the gluing process detailed in Step #13, one drop at a time. **N.B. Do Not let the cyanoacrylate infiltrate the point or run onto the tang.**
15. After the cyanoacrylate sets -- usually between 15 seconds and 2 minutes -- remove the bending jig, and "reverse" bend the blade to find any spots which are not glued.
16. Reassemble the point, inserting the pressure spring, the tip, and the tip screws. This step may be done before Step #11 if there is the possibility of point parts becoming lost or if a number of blades, with different tips, are being rewired at the same time.
17. Retape the top 15cm (the length of a US$ bill) of the foible, install a piece of insulating tubing (aka "spaghetti") over the wire, strip the insulation completely from the blade wire to be connected to the guard socket, and reassemble the foil.

Epee Wires - Stripping and Replacing

1. Disassemble the epee. If a new blade is to be wired, skip Steps #2-#4, and thoroughly degrease the blade groove using acetone or other degreaser.
2. Strip as much wire from the blade groove as possible by pulling the wire from the tang end.

3. Remove the complete point by unscrewing it. This usually breaks the wire below the barrel. Continue stripping the wire by pulling from the tip end.
4. Remove any remnants of wire and glue from the blade groove, using a sharpened hacksaw blade, a Dremel rotary tool with a cut-off disc, a solvent (usually acetone), or heat.
5. Disassemble the point, removing the tip screws, the tip, and the pressure spring; then remove the insulated contact block from the barrel by pushing it toward the mouth of the barrel using a small screwdriver or other punch. Place all the removed parts on a small magnet to prevent their loss.
6. To rewire the blade, first gently unwrap the new epee wire, and straighten any kinks either wire may have.
7. Thread about 2" of the two wires through the barrel, and screw the barrel onto the blade, being careful not to pinch or cut the wires in the blade groove. The use of Teflon plumbing sealant tape or medium strength Loctite on the blade threads will help prevent the barrel from loosening.
8. Tighten the barrel using a 6mm wrench or pliers, being careful not to deform or split the barrel by over-tightening it.
9. Gently pull the two wires through the barrel, taking care not to strip the insulation from either wire.
10. Center the contact block in the barrel, and while maintaining a gentle tension on the wires, use a point setter to seat the contact block at the bottom of the barrel.
11. Using a continuity checker connected to the stripped ends of the two wires, check for any shorts (*e.g.*, between the two wires, and either wire and the blade). There should be no continuity.
12. Straighten both wires and align them in the blade groove under slight tension. Wrap the excess wire around the tang to hold the tension.
13. Place the blade in a blade bending jig, creating a 8"-10 " arc, to hold the wires at the bottom of the blade groove.
14. Holding the blade by the tang, apply a thin cyanoacrylate glue to the blade groove one drop at a time until the wicking glue reaches the middle of the blade.. With certain blades (*e.g.*, Leon Paul, Vniti) you may have to depress the wires immediately in front of the tang to get contact between the blade and the blade wires. Wiping the blade groove with a tissue will remove any excess glue and hasten
15. Holding the blade vertically by the tip, apply one drop of thin cyanoacrylate glue to the blade groove ½" behind the point -- the glue will wick upward toward the point -- letting it wick toward the center of the blade. Continue applying the cyanoacrylate, one drop at a time, until the glue reaches the middle of the blade. **N.B. Do Not let the cyanoacrylate infiltrate the point or run onto the tang.**
16. When the cyanoacrylate is dry -- the drying time is between 15 seconds and 2 minutes -- remove the blade bending jig, and "reverse" bend the blade to see if there are any places which were not glued.
17. Reassemble the point, testing both the residual travel and the pressure spring before inserting the tip screws.
18. Install pieces of insulating tubing (aka "spaghetti") on each of the wires before reassembling the epee.

Foils -- Pre-Competition Check

In checking foils for competitions, the use of an ohmmeter capable of reading 2 ohms is essential. If a competition foil has a steady state of less than 2 ohms electrical resistance, it probably doesn't need a lot of checking.

1. Check the physical condition of the foil, checking for rust, a loose handle, a loose point, sections of blade wire which have come unglued (revealed by reverse bending of the blade).
2. Check that all electrical connections are tight.

3. Check the tip screws. On collar-type tips, be sure the screws are tight; on flange-type tips, make sure the screws are not locked tight.
4. Check that the pressure spring will reset a 500 gram weight. Adding extra weight, via a few coins, can add extra insurance.
5. Check that the foible tape is not shredded.

Epees -- Pre-Competition Check

In checking epees for competitions, the use of an ohmmeter capable of reading 2 ohms is essential. If a competition epee has a steady state of less than 2 ohms electrical resistance when the tip is depressed, it probably doesn't need a lot of maintenance.

1. Check the physical condition of the epee, checking for rust, a loose handle, a loose point, sections of blade wire which have become unglued (revealed by reverse bending of the blade).
2. Check that all electrical connections are tight
3. Check that the pressure spring will reset a 750 gram weight. Adding extra weight, via adding a few coins, can add extra insurance.
4. Check that the epee point will pass both the 1.5mm and 0.50mm epee gauge tests.
5. Check the tightness of the two (2) tip screws. If necessary, adjust the travel by shortening or lengthening the contact spring (see "Removing and Adjusting Epee Points," Step #2 above). **N.B.** Do not cut the contact spring to shorten it.
6. Check the tightness of the barrel on the blade. If the barrel is loose, be very careful tightening it -- tighten it slowly -- excessive zeal can break the insulation of the wires creating a short, or breaking the wires.
7. Continuity should be checked between each wire and the guard, and between the two wires. Continuity between either wire and the guard or blade, or between the two wires, when the tip is NOT depressed, indicates a short. There should only be continuity when the tip is depressed..
8. The guard should be cleaned thoroughly,
9. Check that the "spaghetti" insulating tubing, insulating each of the wires, does not protrude in front of the guard, and that it covers each of the wires from the guard to the connections on the guard sockets. .There should be NO tape covering the wires inside the guard.
10. If the insulated wires go through the guard socket, check that each goes through its own hole. Otherwise, the wires should be outside the guard socket, between the guard socket and the guard itself
11. Check that the pad will NOT allow the fencer's fingers easy access to the wires

Sabres -- Pre-Competition Check

1. Check the physical condition of the sabre, making sure all the physical and electrical connections are tight, and that there is no rust present on the blade and guard.
2. Using an ohmmeter, check that the sabre's overall electrical resistance is below 2.0 ohms.
3. Check the dimensions of the blade's cutting edge and tip, using a sabre gauge.
4. Check all the insulation, especially the pommel. **N.B.** There should be NO tang protruding beyond the pommel.